

Iterative Framework for Creep Prediction from Partial Data using the Wilshire-Cano-Stewart Model

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Motivation & Objective



Methodology



Results



Conclusion

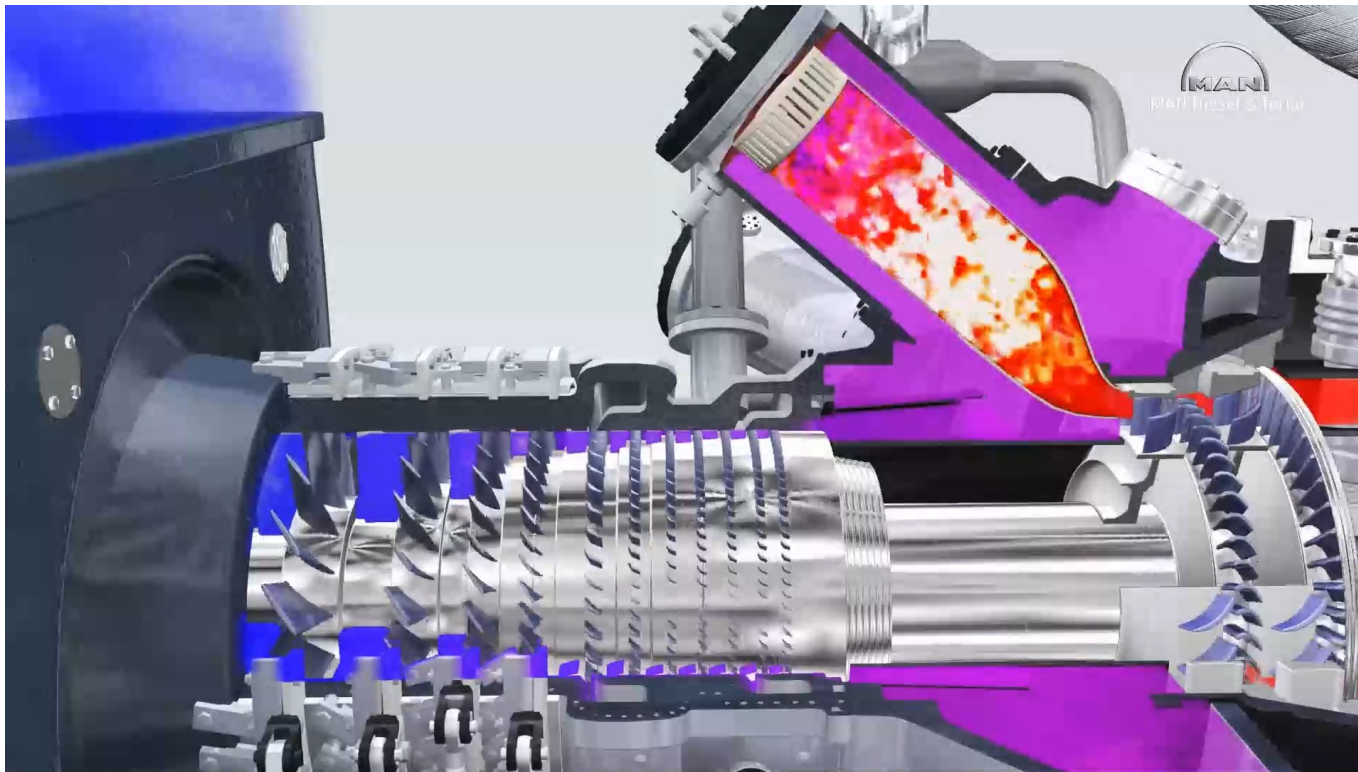
Motivation & Objective



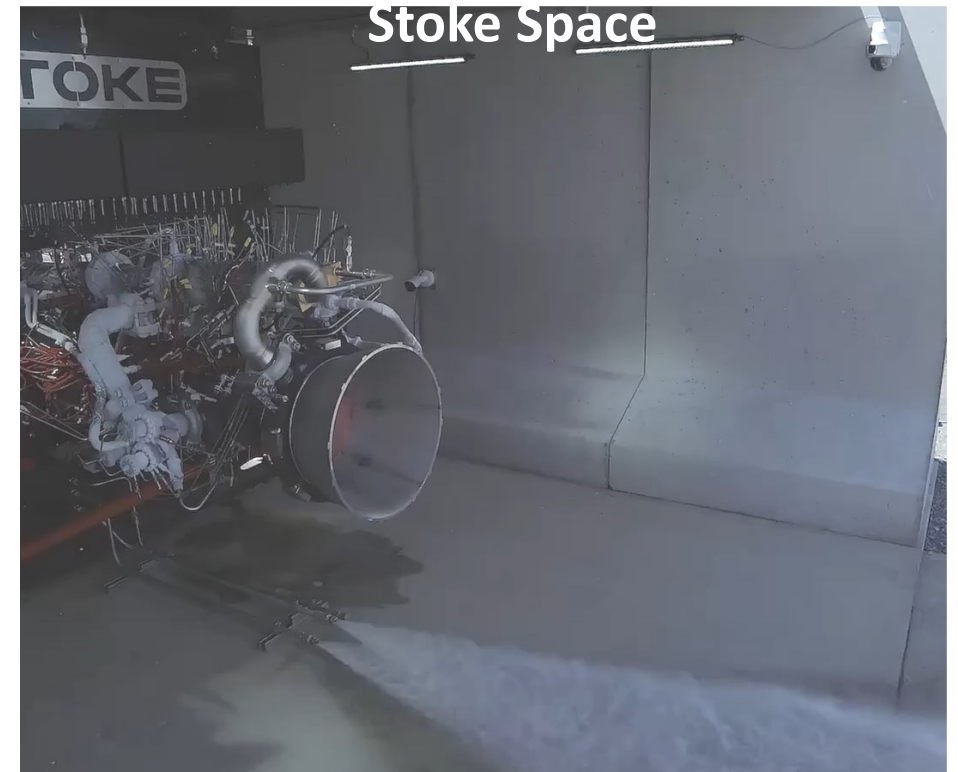
Extreme Environments

- Sustained load + extreme temperature $\rightarrow$ time dependent deformation (creep) limits component life

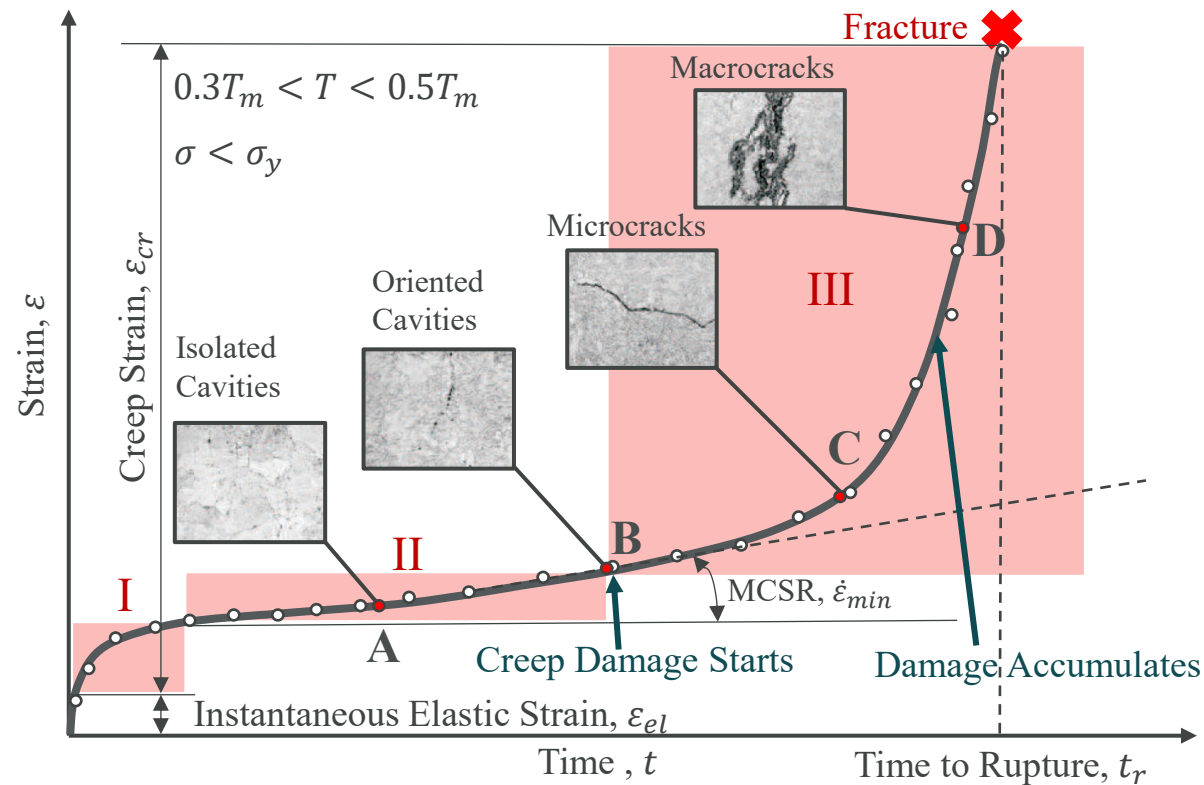
Industrial Gas Turbine



Rocket Engine



Creep and Creep Induced Failure



I - Primary Creep:

Hardening Processes, obstruction of the dislocation movements,
Relaxation Processes, e.g. redistribution of lattice defects

II - Secondary Creep:

Hardening & Softening Processes, are in equilibrium,
Steady-state, or minimum-creep-strain-rate, $\dot{\epsilon}_{min}$

III - Tertiary Creep:

Damage Processes, e.g. formation, growth & coalescence of voids at the grain boundaries, Aging of the microstructure, etc.

A – Observation

B – Observation, fixed inspection intervals

C – Limited service until repair

D – Immediate repair

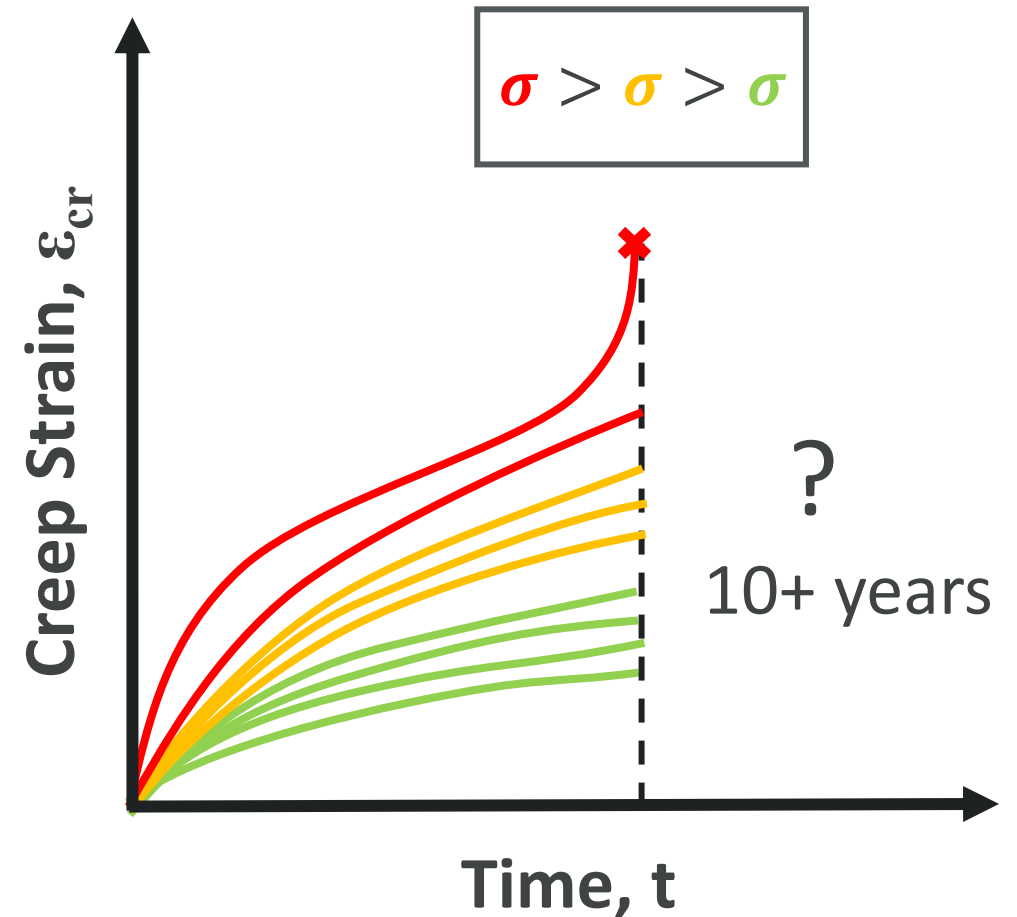


Managing Long-Duration, High-Uncertainty Creep Tests

- Campaigns span years to decades → can we extract early insights?

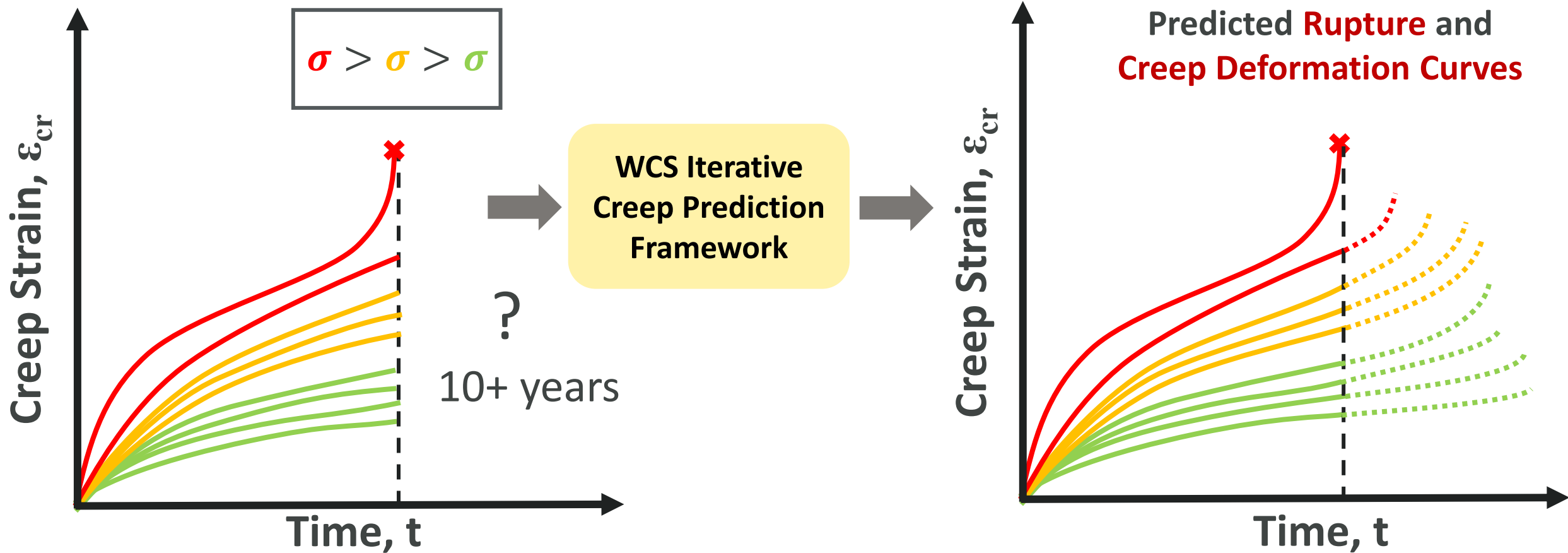


The National Research Institute for Metals in Japan



A Data-Driven Forecasting Tool

- Ongoing tests already contain useful data



Research Objective

- Develop a novel iterative calibration framework for the Wilshire-Cano-Stewart (WCS) model capable of using partial creep data from on-going experiments
- Ground the model's damage evolution physically using Strain Energy Density Dissipation (SED)
- Quantitatively validate the framework's evolving accuracy using Alloy 617 data
- Perform parametric simulations as post-audit validation

Methodology



Constitutive Models

- **McVetty Primary Creep** captures the initial, decelerating stage of creep deformation

Primary Creep Strain

$$\varepsilon_{pr} = G \cdot (1 - e^{-q \cdot t})$$

- **Wilshire-Cano-Stewart (WCS)** is a continuum damage mechanics based constitutive model consist of coupled creep-strain-rate and damage-rate equations

Creep-strain-rate

$$\dot{\varepsilon}_{cr} = \frac{\left[-\frac{\ln(\sigma/\sigma_{TS})}{k_2} \right]^{\frac{1}{v}}}{\exp(Q_c/RT)} \exp(\lambda\omega)$$

Damage-rate equation

$$\dot{\omega} = \frac{[1 - \exp(-\varphi)]}{\varphi} \frac{\exp(-Q_c/RT)}{\left[-\frac{\ln(\sigma/\sigma_{TS})}{k_1} \right]^{\frac{1}{u}}} \exp(\varphi\omega)$$

Material Constant	Behavior
R	Molar Gas Constant
T	Absolute Temperature
σ_{TS}	Ultimate Tensile Strength
G	Max Primary Creep Strain
q	Primary Creep Exhaustion Rate
Q_c	Apparent Activation Energy
k_1, u, k_2, v	Wilshire Model Constants
ϕ	Damage Trajectory Constant
λ	Creep Strain Trajectory Constant
ω	Internal State Variable, Damage

Wilshire-Cano-Stewart (WCS) Model

Rate Based

$$\dot{\epsilon}_{cr} = \frac{\left[-\frac{\ln(\sigma/\sigma_{TS})}{k_2} \right]^{\frac{1}{v}}}{\exp(Q_c/RT)} \exp(\lambda\omega)$$



Closed Form

$$\epsilon(t) = \dot{\epsilon}_{min} \left(\frac{\phi \left((t - t_r)e^\phi - t \right) \left(\frac{t(e^{-\phi} - 1)}{t_r} + 1 \right)^{\frac{-\lambda}{\phi}} + t_r \phi e^\phi}{(\phi - \lambda)(e^\phi - 1)} \right) + \epsilon_{pr}$$

Secondary + Tertiary Creep

Primary Creep

$$\dot{\omega} = \frac{[1 - \exp(-\phi)]}{\phi} \frac{\exp(-Q_c/RT)}{\left[-\frac{\ln(\sigma/\sigma_{TS})}{k_1} \right]^{\frac{1}{u}}} \exp(\phi\omega)$$



$$\omega(t) = \frac{1}{-\phi} \ln \left[1 - [\exp(-\phi)] \frac{t}{t_r} \right]$$

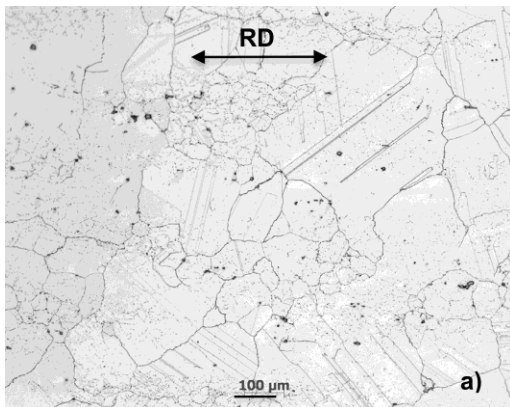
Material: Alloy 617

Alloy 617 is Ni-Cr-Mo-Co based material

Strengthened by solid solution hardening from:

- Chromium
- Molybdenum
- Cobalt
- Intra- and inter-granular carbide precipitates

Applications: Gas turbines, high-temperature nuclear reactors



Mechanical Properties

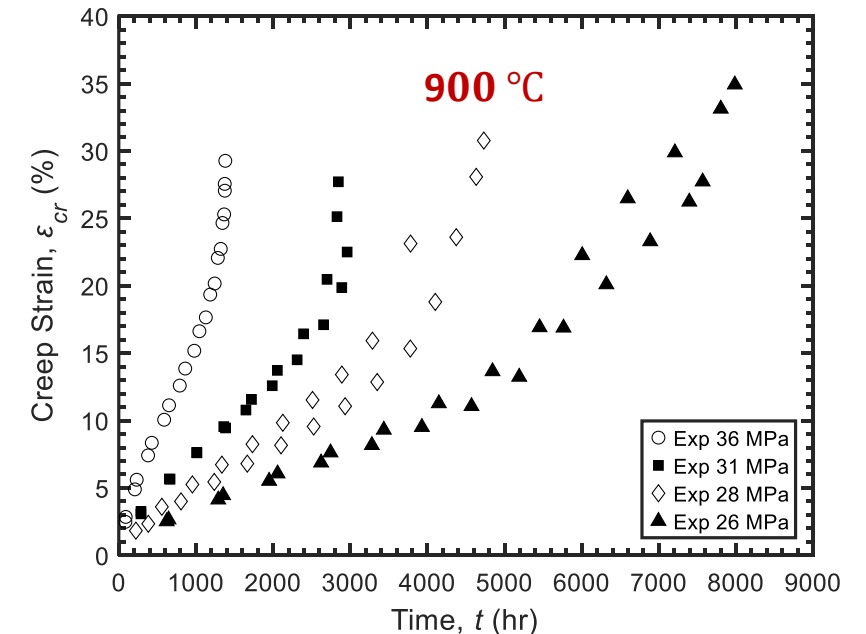
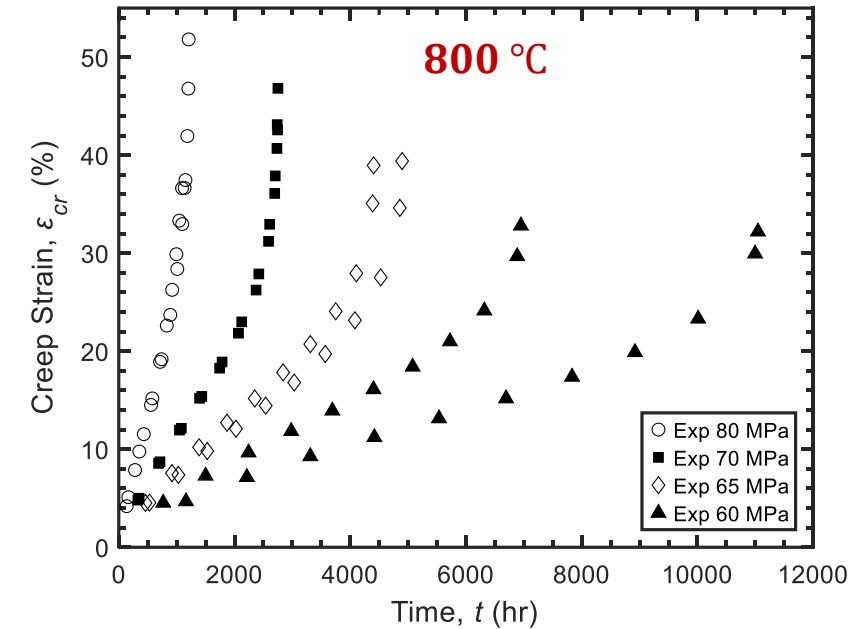
Temp. (°C)	Yield Strength (MPa)	Tensile Strength (MPa)
800	189	290
900	163	166

Chemical Composition

Element	Wt. %	Nominal Wt. %
Ni	54.1	44.5 min
Cr	22.2	20-24
Co	11.6	10-15
Mo	8.6	8-10
Fe	1.6	3 max
Mn	0.1	1 max
Al	1.1	0.8-1.5
C	0.05	0.05-0.15
Cu	0.04	0.5 max
Si	0.1	1 max
S	<0.002	0.015 max
Ti	0.4	0.6 max
B	<0.001	0.006 max

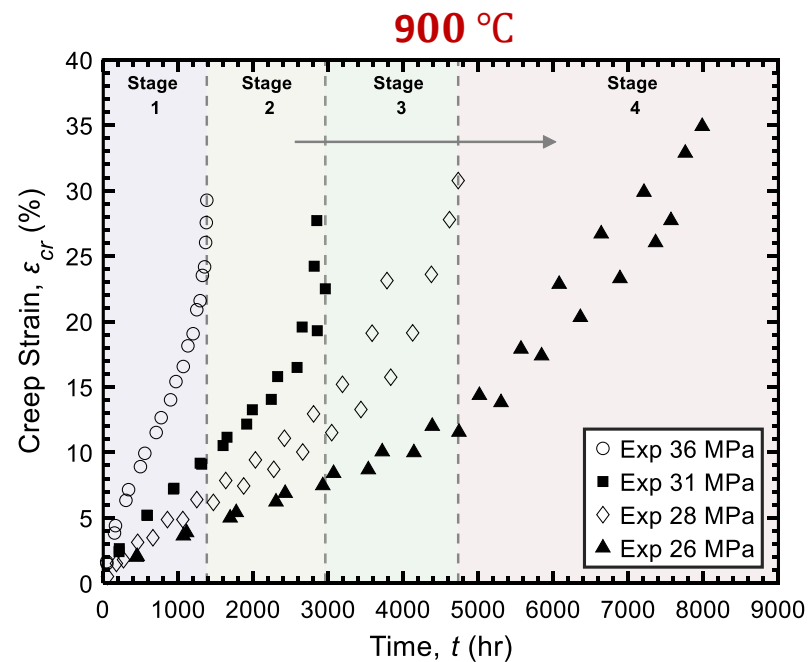
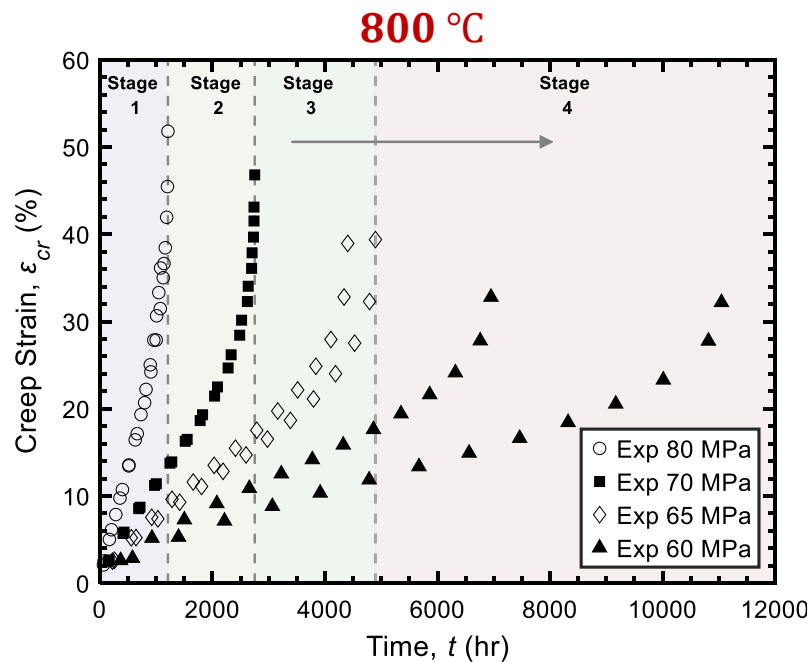
J. Wright, Draft ASME Boiler and Pressure Vessel Code Case for Use of Alloy 617 for Class A Elevated Temperature Service Construction, INL/EXT-15-36305, Idaho National Laboratory, 2015.

Creep Deformation

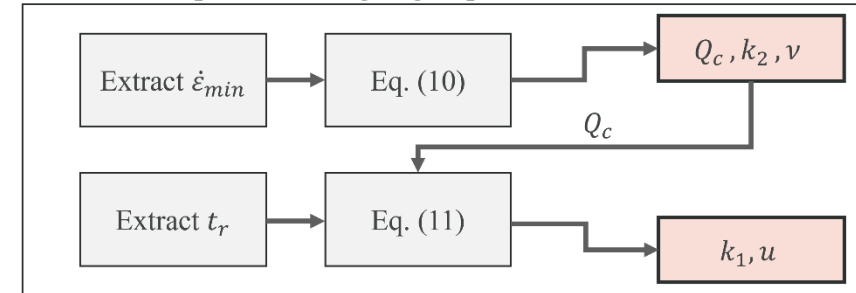


Iterative Calibration Framework

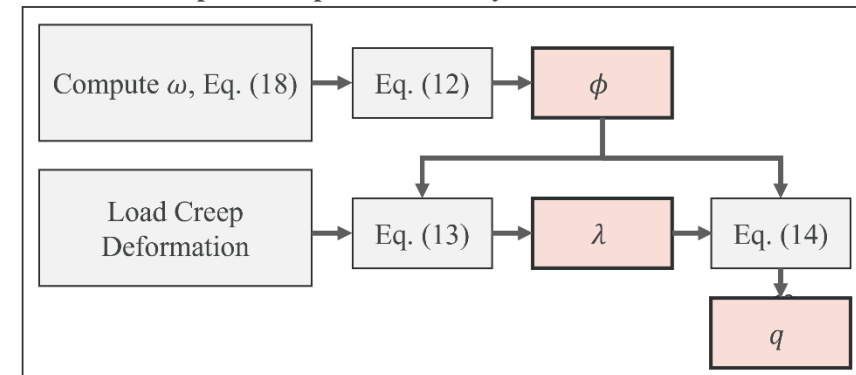
- Use all information available from ongoing tests to improve calibration, fit, and predictions
- Phase 1 — Wilshire eqs $\rightarrow$ MCSR & stress rupture (partial data OK)
- Phase 2 — McVetty + WCS $\rightarrow$ deformation (completed tests)
- Calibration scheme re-runs at completion of each stage



Phase 1: Completed & On-going Experiments



Phase 2: Completed Experiments Only

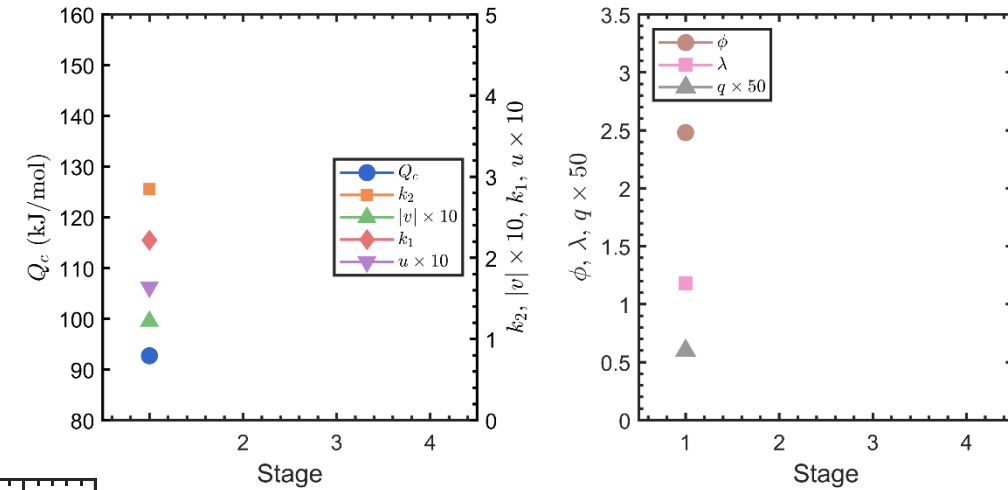
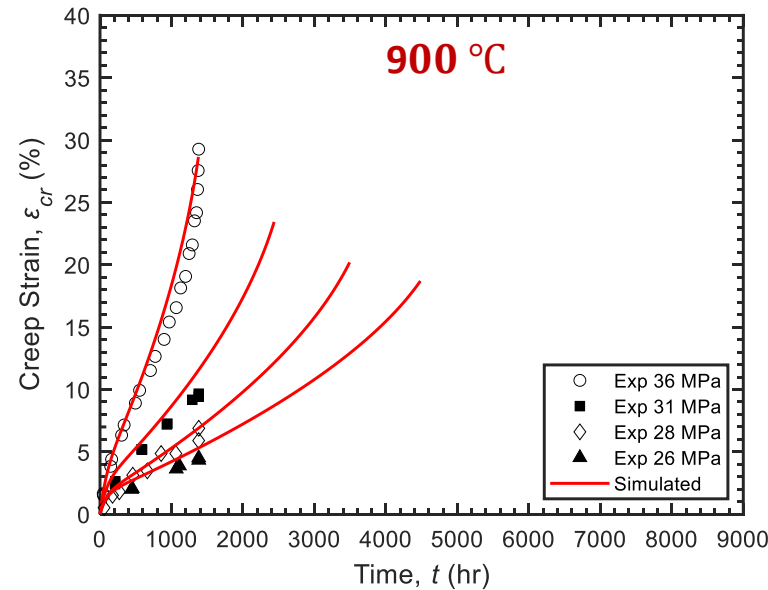
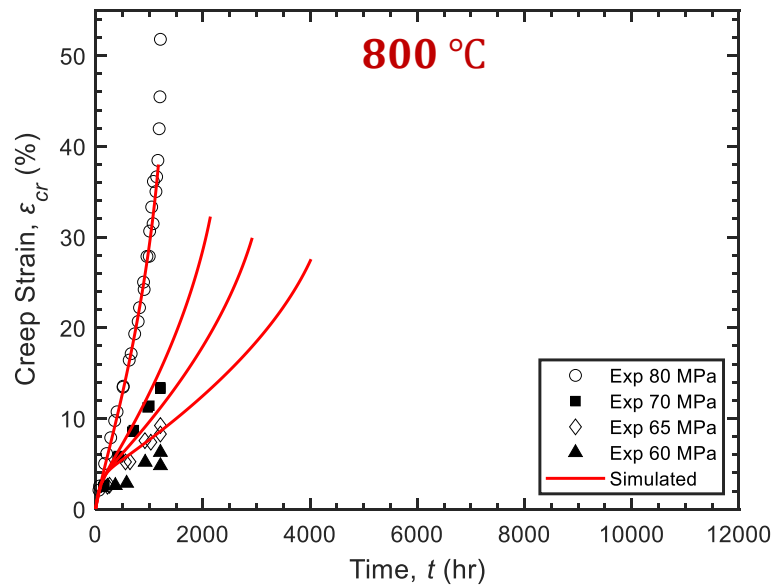


Results



Creep Strain Predictions – Stage 1

- Short term, high stress data only
- On-going test predicted to rupture prematurely

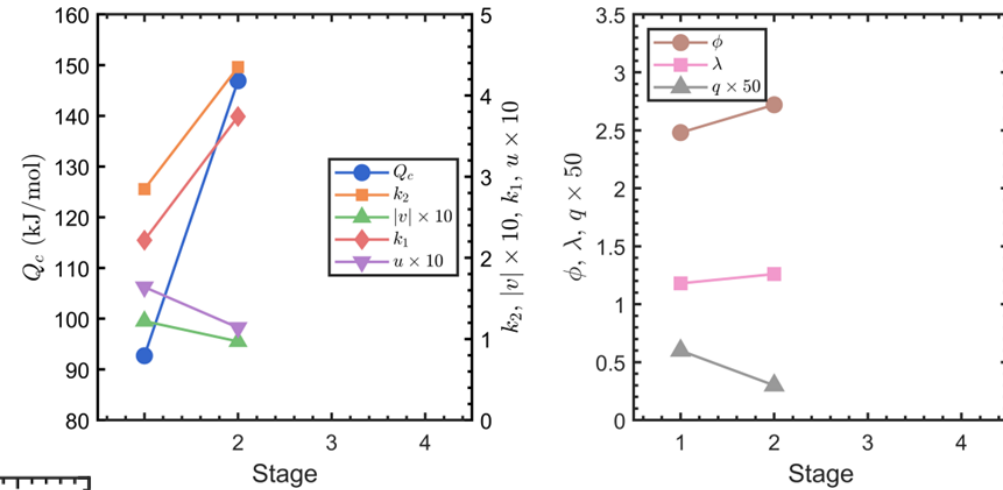
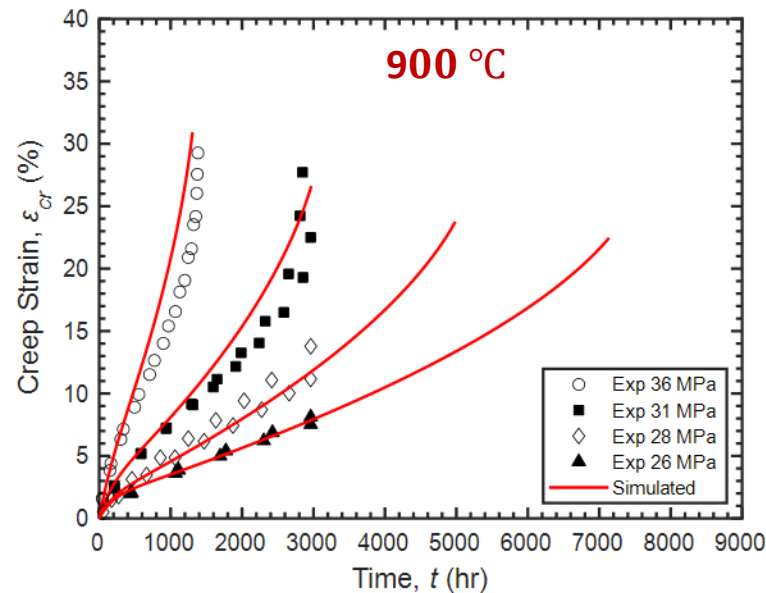
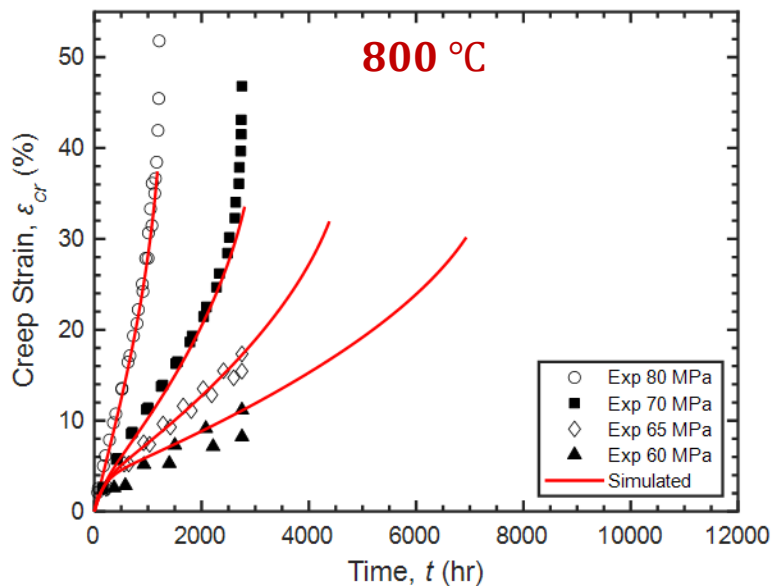


WCS Model Constants Across Calibration Stages

Stage	Q_c kJ/mol	k_2 hr ^{v}	v unitless	k_1 hr ^{$-u$}	u	ϕ unitless	λ unitless	q t^{-1}
1	92.7	2.847	-0.122	2.217	0.164	2.48	1.18	0.012

Creep Strain Predictions – Stage 2

- All test reach MCSR → activation energy locks in
- Predictions land within experimental scatter



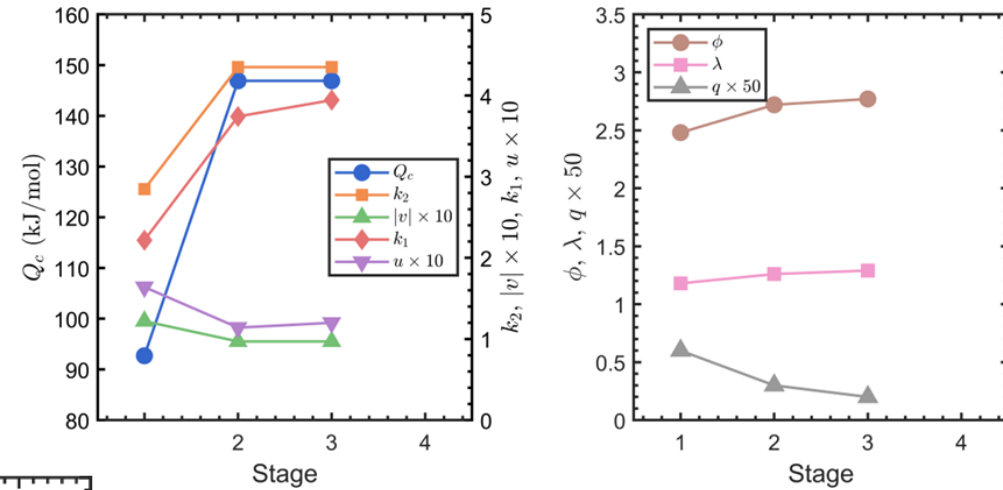
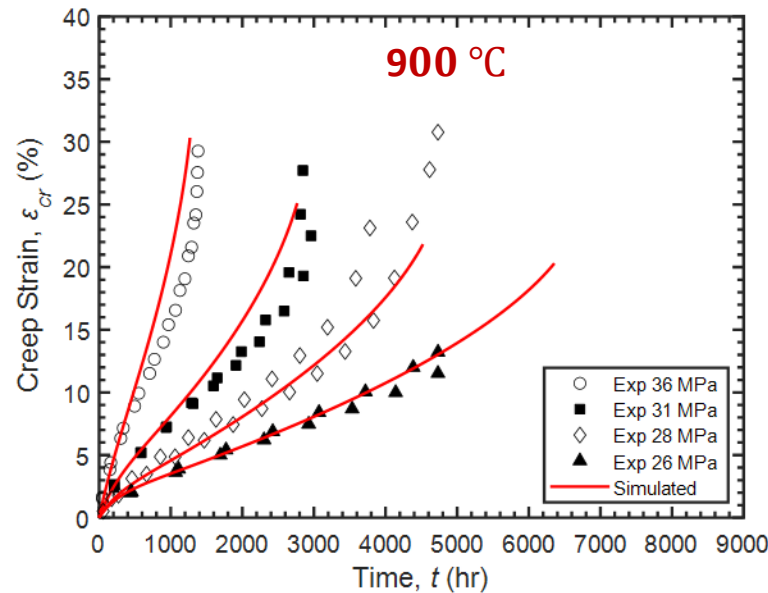
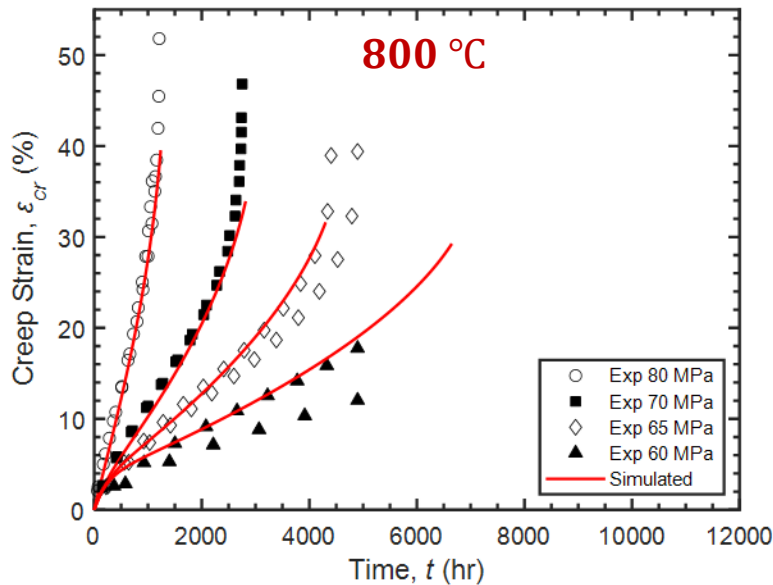
WCS Model Constants Across Calibration Stages

Stage	Q_c kJ/mol	k_2 hr^v	v unitless	k_1 hr^{-u}	u	ϕ unitless	λ unitless	q t^{-1}
1	92.7	2.847	-0.122	2.217	0.164	2.48	1.18	0.012
2	146.9	4.350	-0.097	3.742	0.114	2.72	1.26	0.006



Creep Strain Predictions – Stage 3

- WCS model constants stop moving
- The stability is a good sign



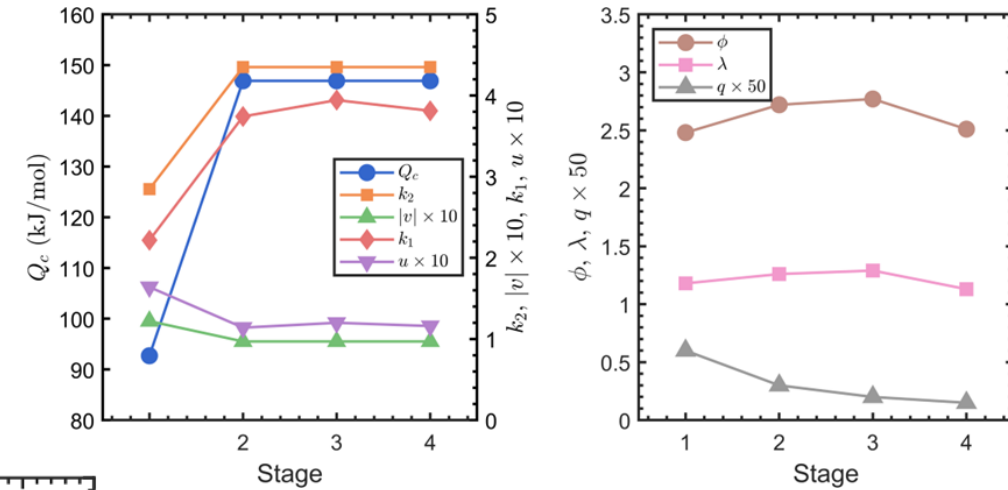
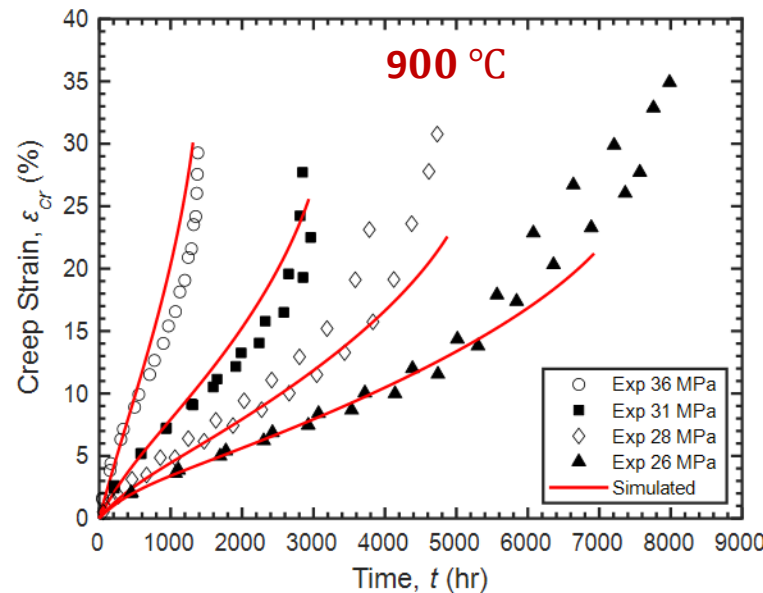
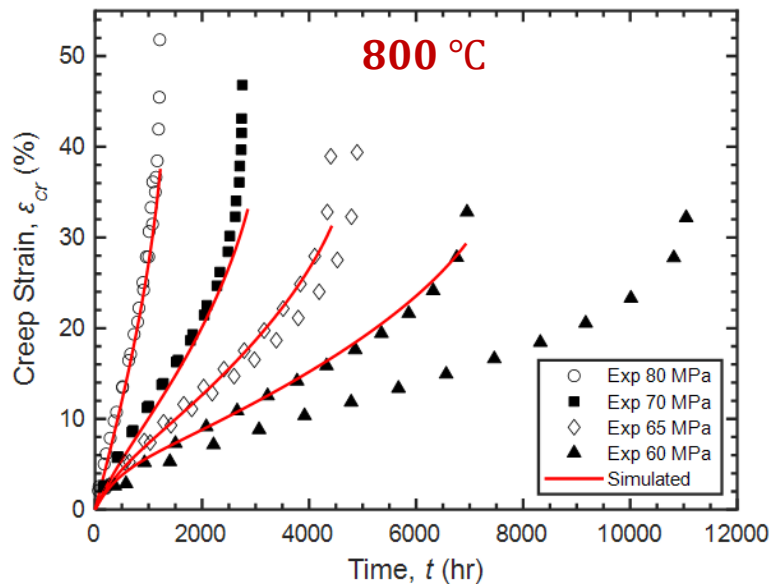
WCS Model Constants Across Calibration Stages

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1	92.7	2.847	-0.122	2.217	0.164	2.48	1.18	0.012
2	146.9	4.350	-0.097	3.742	0.114	2.72	1.26	0.006
3				3.943	0.120	2.77	1.29	0.004



Creep Strain Predictions – Stage 4 (Validation)

- Stage 2 forecast confirmed, ~4,000-5,000 frame-hours of foresight



WCS Model Constants Across Calibration Stages

Stage	Q_c kJ/mol	k_2 hr v	v unitless	k_1 hr $^{-u}$	u	ϕ unitless	λ unitless	q t^{-1}
1	92.7	2.847	-0.122	2.217	0.164	2.48	1.18	0.012
★ 2				3.742	0.114	2.72	1.26	0.006
3	146.9	4.350	-0.097	3.943	0.120	2.77	1.29	0.004
4				3.812	0.116	2.51	1.13	0.003

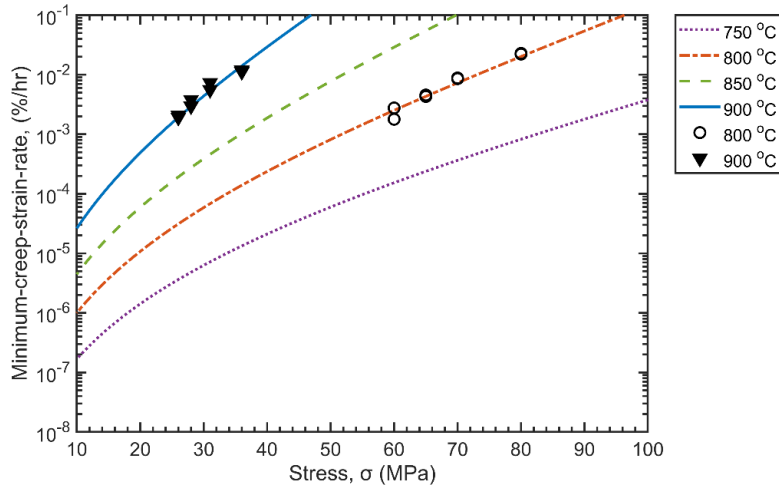
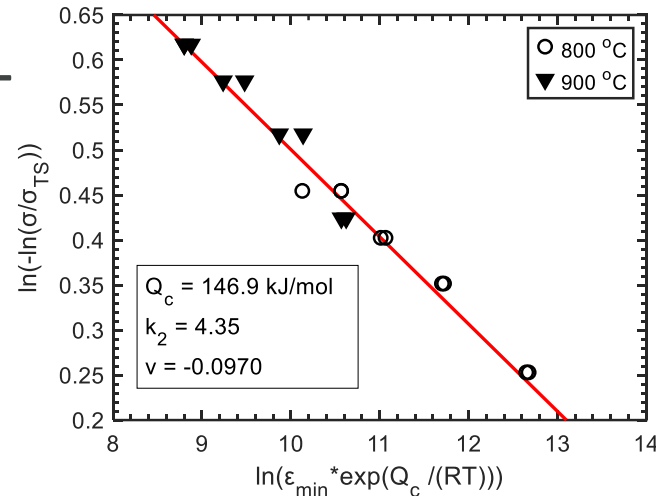


Wilshire Plots – Stage 4 Constants

- Well-tuned Q_c collapses both isotherms onto a linear fit

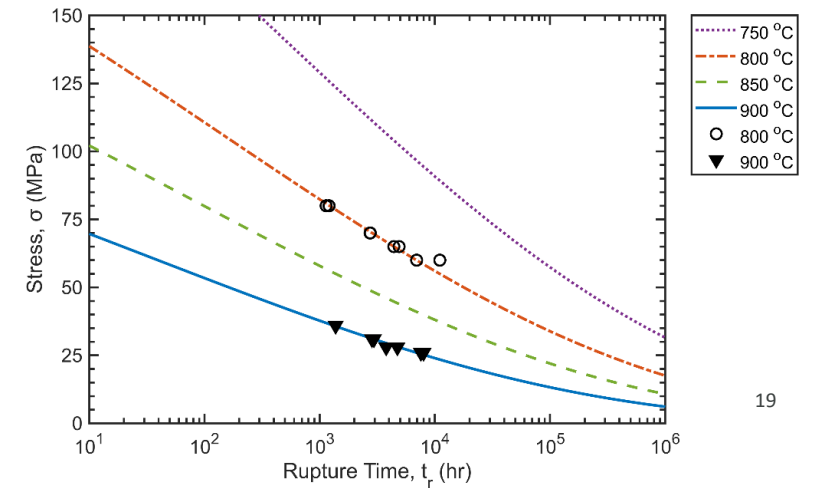
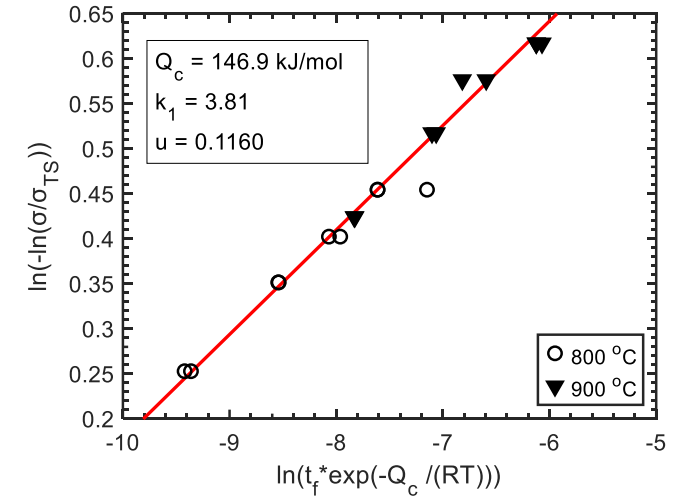
Minimum-Creep-Strain-Rate

$$\dot{\epsilon}_{min} = \frac{\left[-\frac{\ln(\sigma/\sigma_{TS})}{k_2} \right]^{\frac{1}{v}}}{\exp(Q_c/RT)}$$



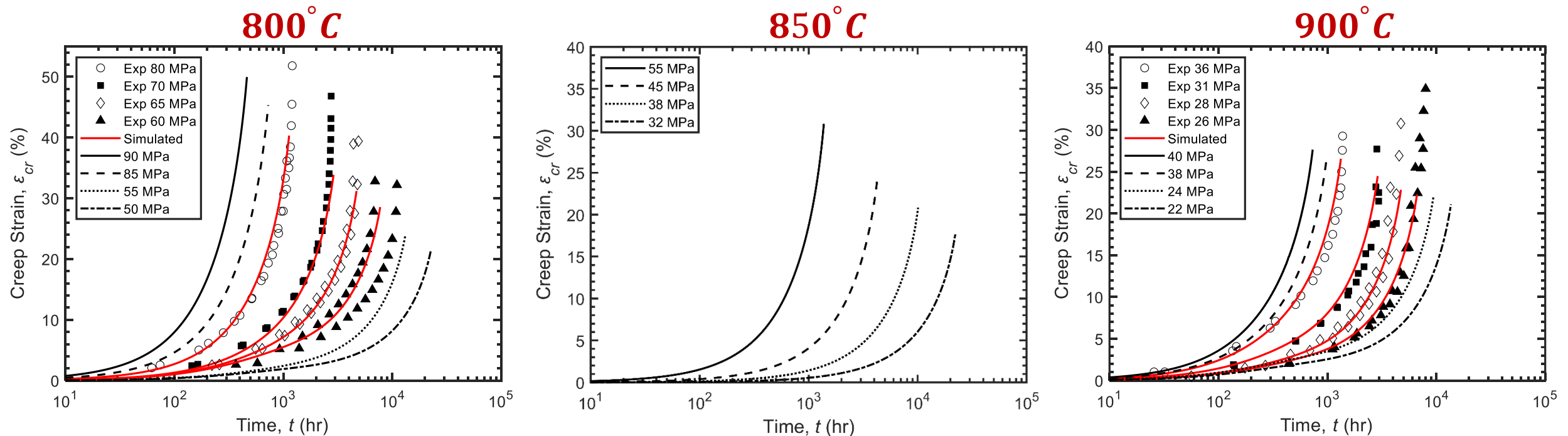
Stress Rupture

$$t_r = \frac{\left[-\frac{\ln(\sigma/\sigma_{TS})}{k_1} \right]^{\frac{1}{u}}}{\exp(-Q_c/RT)}$$



Post Audit Validation – Stage 4 Constants

- Single set of WCS constants able to interpolate and extrapolate creep strain across isotherms
- Predictions are in good agreement with experimental data



Discussion

- Alloy 709
 - ASME Sec. III, Div. 5 qualification in progress → 10+ years to complete
 - Design parameters require full creep curves → Includes 62 condition creep-rupture matrix
- Heat-to-heat go/no-go
 - Code Case requires ≥ 3 commercial heats
 - Forecast from partial data stronger/weaker verdict

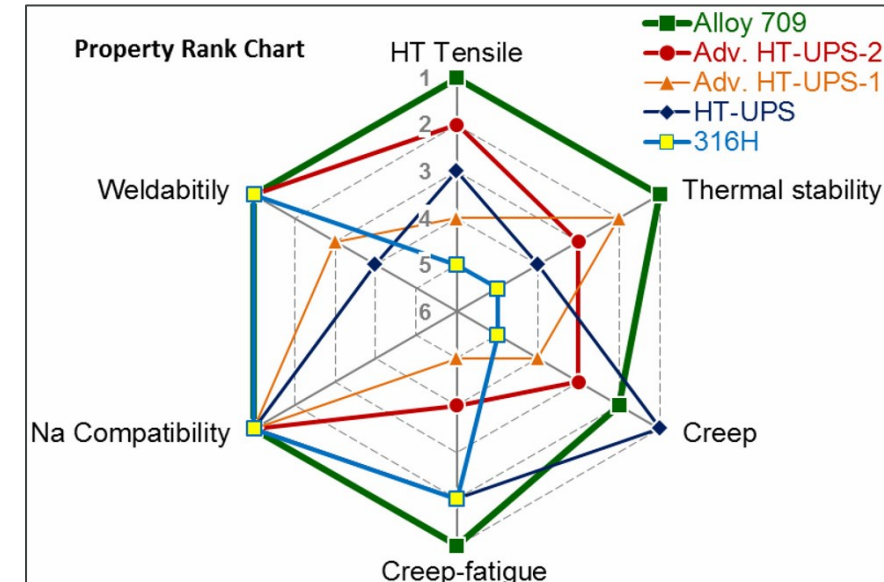


Fig: Property rank chart for austenitic alloys

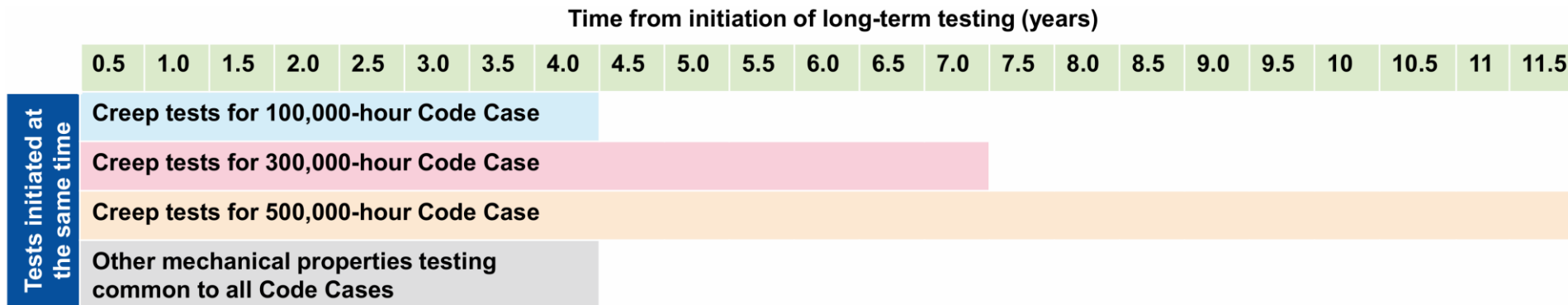
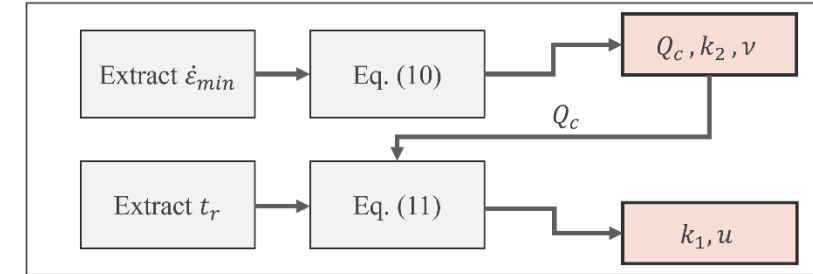


Fig: “Staged” approach for the Code qualification of Alloy 709

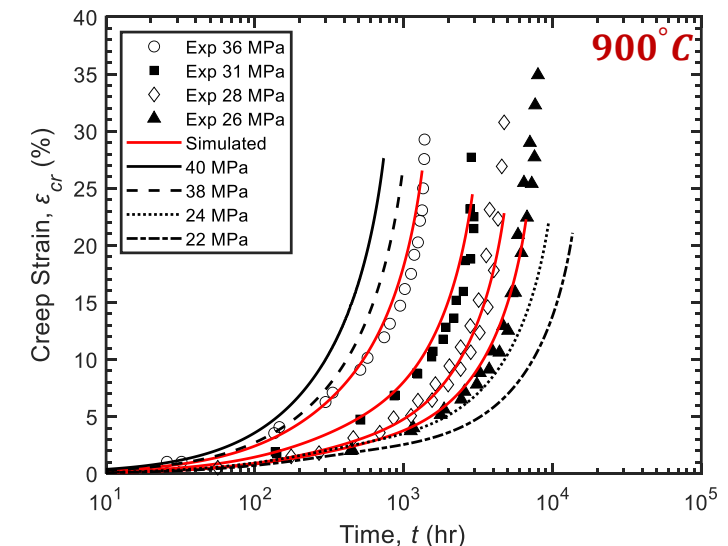
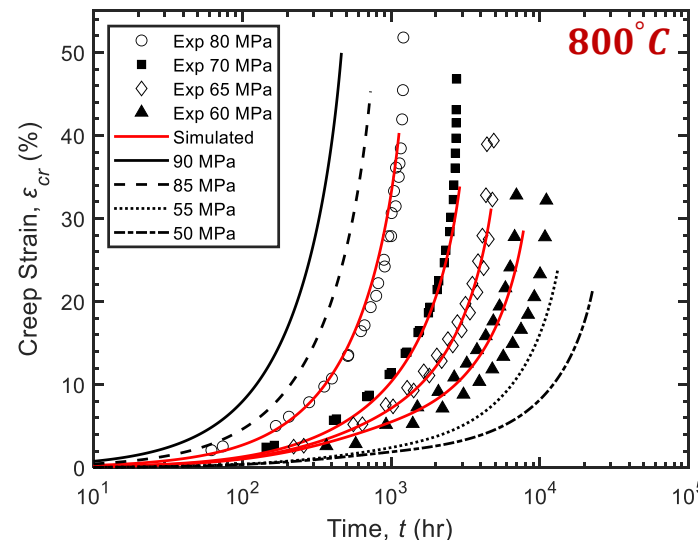
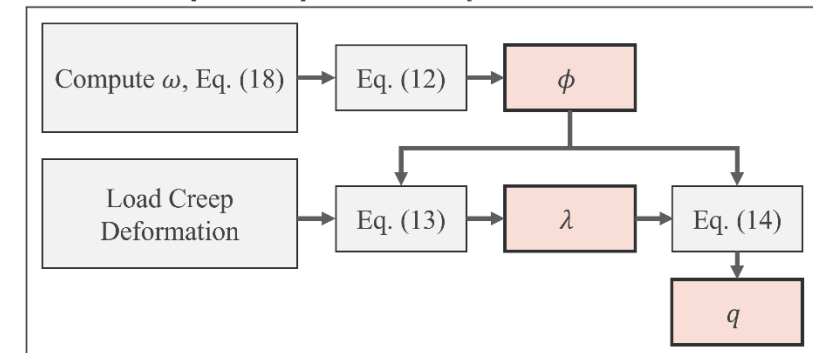
Conclusion

- Introduced an iterative calibration framework using Wilshire-Cano-Stewart (WCS) model
- Complete creep predictions can be made from partial experimental data
- Predictions improves as more data is collected
- Validated the framework using Alloy 617 data

Phase 1: Completed & On-going Experiments



Phase 2: Completed Experiments Only



Thank You...



Rafi Khan

The Ohio State University



Kojo Benefo

The Ohio State University



Dr. Calvin M Stewart

The Ohio State University

MATERIALS AT EXTREMES



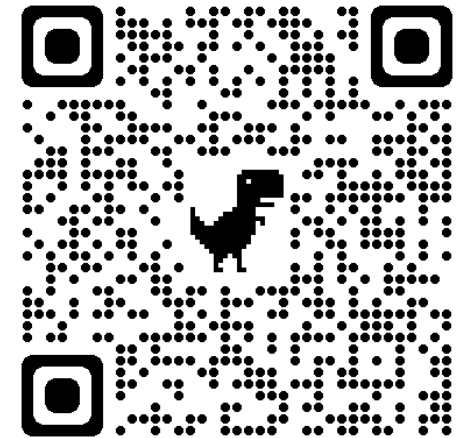
The Ohio State University

Department of Mechanical and Aerospace Engineering
N350 Scott Laboratory
201 W 19th Avenue, Columbus, OH 43210

Materials at Extremes (MATX)

<https://mae.osu.edu/MATX>

Principal Investigator: Calvin M. Stewart



MATX Labs



THE OHIO STATE UNIVERSITY
COLLEGE OF ENGINEERING

MATX

MATERIALS AT EXTREMES

focuses on the advanced manufacturing, testing & characterization, and theoretical mechanics of materials subject to extreme environments.

Extra Slides



Strain Energy Density Dissipation (SED)

In the **Strain Energy Density (SED)** approach, assume that a material has a finite amount of energy it can dissipate through plastic deformation until reaching rupture

$$W_f = \int_0^{t_r} \sigma(t) \dot{\epsilon} dt$$

SED expressed as cumulative experimental damage

$$\omega = \frac{W}{W_f}, \quad 0 \leq \omega \leq 1 \quad \text{where } 0 \text{ (undamaged) and } 1 \text{ (rupture)}$$

A single ϕ is determined to satisfy the damage evolution described using SED

$$\omega(t) = -\frac{1}{\phi} \ln \left[1 - [1 - \exp(-\phi)] \frac{t}{t_r} \right] \quad \text{where } \phi \geq 0$$

Advantages

- Removes stress and temperature sensitivity
- Provides an experimentally realistic damage to improve modeling repeatability and accuracy

